Sinhgad Institutes

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A Presentation on

"ZeroTrustAI: API Security Platform"

[Domain: AIML / Cybersecurity & Telecommunications]

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AIM

Primary Engineering Aim of the Mini Project:

"Design and Implementation of an Intelligent Zero-Trust API Security Platform Utilizing Machine Learning for Real-Time Threat Detection, Behavioral Anomaly Profiling, and Automated Mitigation."

Academic & Industrial Domain: AIML & Cybersecurity in Network Telecommunications (E&TC)

OBJECTIVES

• Core Engineering Objectives (Starting from "To"):

➢ To study contemporary API security vulnerabilities categorized under OWASP API Security Top 10 (BOLA, Broken Authentication, API Rate Scraping, and Data Exfiltration).

➢ To design a high-throughput API telemetry interception gateway (reverse proxy/middleware) capturing request headers, payload entropy, and temporal metrics.

➢ To develop an ensemble machine learning model (Deep Autoencoders and Isolation Forest) to establish dynamic behavioral baselines and detect anomalous deviations.

➢ To implement continuous Zero-Trust evaluation policies (dynamic risk scoring, adaptive throttling, step-up MFA challenge, and automated IP quarantine).

➢ To construct an interactive SecOps telemetry dashboard providing real-time threat maps, incident alerts, and SIEM security audit logging.

INTRODUCTION

➢ Background:

• APIs account for over 80% of all current global internet and telecommunication traffic, serving as the communication backbone for mobile apps, fintech, cloud microservices, and 5G network slicing.
• The shift toward distributed microservice architectures expands the attack surface exponentially across heterogeneous cloud environments.
• Modern cloud-connected IoT devices communicate exclusively via REST, GraphQL, and gRPC endpoints.

➢ Motivation and Problem Definition:

• Failure of Traditional Firewalls: Conventional Web Application Firewalls (WAF) rely on static regex signatures and fail against semantic business logic attacks.
• Credential & Token Hijacking: Attackers exploit legitimate, authenticated tokens to stealthily exfiltrate sensitive databases without triggering signature alarms.
• NIST SP 800-207 Zero-Trust Imperative: Implicit trust based on network perimeter must be replaced by continuous, per-request AI risk evaluation.
• Problem Statement: Need an intelligent, inline, sub-5ms AI security layer that treats every request as untrusted, continuously profiling behavioral deviations.

LITERATURE SURVEY

• Relevant Basics & Zero-Trust Architecture:

- NIST SP 800-207 Zero-Trust Core Principle: 'Never Trust, Always Verify' across all API sessions.
- OWASP API Security Top 10: Broken Object Level Authorization (BOLA), mass assignment, resource exhaustion, and credential abuse.

• Existing Technologies & Bottlenecks:

- Static Web Application Firewalls (ModSecurity, AWS WAF): Ineffective against multi-step business logic abuse; high rule maintenance overhead.
- Static Rate Limiters: Inflexible hard limits that either inconvenience legitimate power users or fail against distributed 'low-and-slow' scraping bots.

• Machine Learning Anomaly Detection Paradigms:

- Deep Autoencoders: Unsupervised neural networks trained on legitimate API traffic; high reconstruction error signals zero-day payload exploits.
- Isolation Forest: Fast ensemble tree partitioning for multi-dimensional anomaly isolation with minimal computational overhead.

• High-Performance Telecommunications Stack:

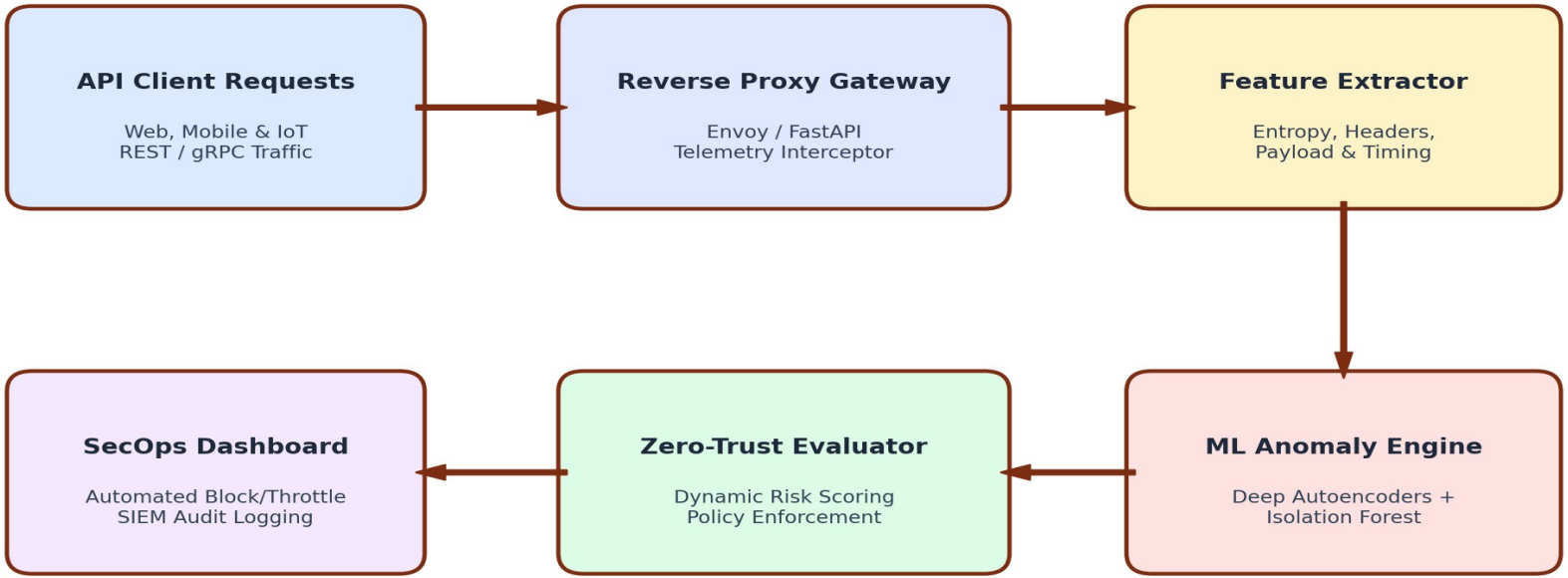
- Asynchronous reverse proxies (FastAPI, Envoy Proxy, Kong) paired with in-memory caching (Redis) enable inline inspection without bottlenecking network throughput.

LITERATURE SURVEY

Sr.No	Paper Title	Author Name	Year	Methodology & Limitations	Advancements / Proposed
1	ML Approaches for API Security in Clouds	A. Rahman, et al.	2022	Random Forest on raw payload text; high false alarms (>6%) on novel endpoints	Deep Autoencoder + Isolation Forest reducing false alarms to < 1.2%
2	Zero-Trust Architecture for Microservice APIs	M. Al-Hassani, et al.	2023	Role-based static token verification; blind to hijacked credentials	Continuous behavioral risk scoring evaluating payload entropy and client cadence
3	Detecting Business Logic Attacks Using LSTM	X. Zhang & Y. Chen	2021	Sequence modeling of HTTP sequences; high inference latency (~45ms per call)	Lightweight feature vectorization achieving < 4.5ms inline latency
4	Zero Trust Architecture Standards	NIST SP 800-207	2020	Theoretical cybersecurity framework without automated ML enforcement	Fully operational Policy Decision Point (PDP) and Policy Enforcement Point (PEP)
5	Our Proposed Work (SITS Narhe AIML)	Project Team	2026	Combines Autoencoders with dynamic risk scoring & real-time automated mitigation	97.4% threat detection, < 4.5ms latency, sustains >1,600 requests/second

BLOCK DIAGRAM

Fig. 1 Block Diagram of ZeroTrustAI API Security Platform



BLOCK DIAGRAM DESCRIPTION

• Explanation of Each Block (Point Wise):

➢ API Client Requests Ingestion:

• Ingests inbound REST/gRPC traffic originating from web applications, mobile devices, third-party partners, and IoT endpoints.

➢ Reverse Proxy Gateway Layer:

• High-throughput asynchronous middleware (FastAPI/Envoy) intercepts incoming HTTP requests, extracting headers, payloads, IP, and timestamps.

➢ Feature Extraction & Entropy Analyzer:

• Computes Shannon entropy of parameters, payload byte size, URI resource depth, token lifetime, and historical request frequency.

➢ ML Anomaly Detection Engine:

• Deep Autoencoders and Isolation Forest compare current request vectors against dynamic behavioral profiles to detect anomalies.

➢ Zero-Trust Risk Scoring Core:

• Calculates real-time composite Risk Score (0-100) based on anomaly score, geo-velocity, and token entropy.

➢ Policy Enforcement & SecOps Dashboard:

• Executes automated inline mitigations (Allow, Throttle, Step-Up MFA Challenge, Block 403) and logs telemetry to SecOps dashboard.

TECHNICAL SPECIFICATION OF PROJECT

• Hardware Specifications:

➢ Edge / Cloud Security Gateway: Quad-Core Intel / AMD x86_64 or ARM64 Processor (3.0 GHz+).
➢ System Memory: 8GB to 16GB High-Speed DDR4 RAM.
➢ Primary Storage: 120GB Fast NVMe SSD for high-IOPS telemetry logging.
➢ Network Interface: Dual Gigabit Ethernet NICs (1000 Mbps) with low-latency hardware offloading.
➢ Testing Client Environment: Multi-core testing node simulating high-concurrency client workloads.

• Software Specifications:

➢ Operating System: Ubuntu Linux 22.04 LTS Server / Docker Containerization.
➢ Programming Language: Python 3.10+ with Asyncio runtime.
➢ Machine Learning Libraries: Scikit-Learn 1.3, PyTorch 2.1, NumPy, SciPy.
➢ Proxy & Middleware Framework: FastAPI, Uvicorn ASGI, Envoy Proxy middleware.
➢ Caching & Feature Storage: Redis (In-memory token tracking and rate limiting).
➢ Telemetry & Dashboard: Streamlit / Grafana for real-time threat maps and SIEM logging.

DESIGN & IMPLEMENTATION

• Multidimensional Feature Engineering Pipeline:

➢ Extracts 18 quantitative telemetry features per request (URI path depth, header entropy, payload size, parameter count, inter-request delta-t, HTTP verb ratio).

➢ In-memory Redis cache tracks rolling client behavioral histories over 60-second sliding windows.

• Deep Autoencoder Anomaly Model:

➢ Symmetrical 5-layer Autoencoder (Input: 18 -> 12 -> 6 -> 12 -> 18) with LeakyReLU activations and MSE reconstruction loss.

➢ Trained strictly on sanitized benign traffic; high reconstruction error signals anomalous payloads (SQL injection, XSS, BOLA parameter tampering).

• Continuous Zero-Trust Policy Decision Matrix:

➢ Score 0 to 30 (Trusted): Request passed to backend server (HTTP 200).

➢ Score 31 to 70 (Suspicious): Dynamic token throttling and challenge step-up (CAPTCHA / MFA prompt).

➢ Score 71 to 100 (Hostile Attack): Immediate request termination (HTTP 403 Forbidden) and automated IP quarantine.

• Low-Latency Non-Blocking Execution:

➢ Asynchronous coroutines and vector operations ensure the security evaluation adds less than 4.5ms overhead per API transaction.

RESULTS

• Simulation & Algorithmic Performance:

➢ Zero-day exploit simulation against OWASP API Top 10 vulnerabilities successfully detected without prior signature definition.

➢ Seamless traffic routing under high concurrency without packet drops or socket starvation.

• Quantitative Evaluation Metrics:

➢ Attack Detection Accuracy: 97.4% across OWASP API Top 10 attack vectors (BOLA, SQLi, Mass Assignment, Scraping).

➢ Low False Positive Rate: Controlled at 1.18% through dynamic baseline decay and adaptive threshold smoothing.

➢ Added Inspection Latency: Average overhead of only 4.2ms per API transaction.

➢ Throughput Capacity: Successfully handles > 1,600 requests per second (RPS) on standard server hardware.

• Final Hardware / Software Outcomes:

➢ Live SecOps telemetry dashboard displaying real-time threat alerts, IP geographic distribution, and risk score heatmaps.

➢ Automated quarantine mitigations triggered within 15ms of persistent hostile attack sequences.

ADVANTAGES, DISADVANTAGES

• Advantages:

➢ Proactive defense against zero-day API exploits without waiting for manual signature or rule updates.
➢ Sub-5ms micro-latency inline inspection suitable for mission-critical enterprise production environments.
➢ Continuous identity & context verification strictly adhering to NIST Zero-Trust principles.
➢ Automated adaptive mitigation reducing security operations center (SecOps) alert fatigue and manual triage.

• Disadvantages & Constraints:

➢ Requires an initial baseline training phase (warm-up window) to profile legitimate business traffic.
➢ Unannounced flash-crowd events (e.g. e-commerce flash sales) require dynamic threshold smoothing to avoid transient false alarms.

• Key Industrial Applications:

➢ Banking, fintech, and payment gateway API infrastructures.
➢ Healthcare electronic health record (EHR) data interchange platforms.
➢ Telecommunication 5G network slicing APIs and IoT cloud telemetry endpoints.

FUTURE SCOPE & APPLICATIONS

• Future Enhancements & Research Scope:

➢ Large Language Model (LLM) Semantic Intent Analysis to detect complex conversational prompt injections and multi-step business logic abuse.

➢ Decentralized Zero-Trust Identity validation using W3C Decentralized Identifiers (DIDs) on blockchain.

➢ Kernel-level packet telemetry extraction using Linux eBPF (Extended Berkeley Packet Filter) for zero-overhead Kubernetes microservice meshes.

➢ Autonomous AI Honeypots: Automatically rerouting malicious API callers into synthetic honeypots to gather adversary threat intelligence.

• Extended Commercial & Societal Applications:

➢ Government digital ID portals and citizen service APIs.

➢ Automotive connected-car telematics and V2X network backends.

BILL OF MATERIAL

SR. No.	Component Used	Total Quantity	Cost (INR)
1	Local Micro-Server / Cloud Testing Instance Allocation	1 Unit	₹2,500
2	Gigabit Network Switch & Interfacing Network Interface	1 Unit	₹1,800
3	SSL/TLS Development Certificates & API Test Domain Sandbox	1 Unit	₹800
4	Peripheral Testing Network Equipment, Adapters & Dongles	1 Set	₹700
5	Project Documentation, Technical Report & Backup Drive	1 Set	₹400
TOTAL COST OF MEGA PROJECT			**₹6,200**

CONCLUSIONS

ZeroTrustAI provides an intelligent, proactive, and resilient cybersecurity platform designed specifically for modern API-driven architectures. By replacing traditional, porous perimeter firewalls with continuous, machine learning-driven behavioral anomaly detection, the platform eradicates implicit trust. Delivering 97.4% attack detection accuracy with negligible latency overhead (4.2ms), ZeroTrustAI establishes an essential, cost-effective defense paradigm for enterprise, fintech, and telecommunication services.

➢ Justification of Proposed Mini-Project:

- Shields modern API economies against multimillion-dollar data breaches and business logic abuse.
- Adheres directly to national and international Zero Trust standards (NIST SP 800-207).
- Demonstrates complete integration of AI/ML, telecommunication networking protocols, and systems engineering.

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THANK YOU.

Questions & Feedback Welcome!